

Communicating with the Entire Population: Machine Translation of Emergency Alerts and Warnings

Jen Wilson*, Vipasha Bansal, Will Lewis

Department of Linguistics
University of Washington

Abstract

It is essential that we are able to communicate with the entirety of an affected population during a crisis, including those who do not speak the dominant language(s) of a region. Commercial Translation tools are tempting solutions to broaden access across languages and can provide a lifeline when no other resources are available. Emergency response managers, however, are operating in the dark when using these tools, especially with respect to their quality and adequacy. In this paper we seek to shed light onto the quality and utility of commercial translation tools, such as Google Translate or ChatGPT, and offer guidance on how and when these tools can be used

1 Introduction

Communication in emergency and disaster response is critical. In fact, “...access to information in disasters is equally important as access to food, water, shelter and medication” (Meier, 2015). It is thus essential that emergency response systems—those responsible for conveying emergency alerts and warnings, including Wireless Emergency Alert (WEA) systems—communicate with the *entirety* of an affected population, not just a subset. Anything short of that puts people’s lives at risk.

However, in a world where people are increasingly mobile, population and linguistic diversity can introduce seemingly insurmountable challenges. Local emergency managers, public information managers, communication managers, and the like, especially in larger municipalities and regions, are confronted not with a population that speaks a handful of languages, but rather a population that speaks upwards of 100 or 200 languages, with no guarantee of bilingualism or multilingualism. If emergency alerts and notifications are distributed in just a few dominant languages, it is

almost guaranteed that some subsets of the population will be left out. This can have devastating consequences for local constituents in time-sensitive crises.

But how does one go about translating messages into dozens or hundreds of languages? A local government usually does not have the budget for hiring a team of translators. Technological solutions may seem like a safe bet, given the importance of getting information disseminated quickly and without interruption. Commercial Machine Translation (MT) tools, such as Google Translate, Microsoft Translator, ChatGPT, Gemini, DeepL, are readily available on all manner of devices and support a large number of languages. This would seem to make using MT in emergency response an obvious choice. Yet, as important as it is to “get the message out” during a crisis, it is equally important that the message be received and understood as intended. If information is translated incorrectly and then distributed, a subset of the population might receive an alert that is at best uninterpretable, or at worst provides dangerously incorrect information. In other words, there is no guarantee that a commercial translation tool will correctly translate any single message into a given language.

The fact that it translates well into one language is no guarantee that it will translate well into another. Further, even for a specific language, there is no guarantee that MT systems will translate equally well across subject domains.

Emergency managers are often working in the dark when it comes to commercially available translation tools. Given the amount of hype and noise around all aspects of Artificial Intelligence (AI)—Machine Translation (MT) being one of many AI tools currently available—it is difficult to discern utility from hype. In this paper, we seek to cast some light on the issue of translation in the emergency response space, especially when

*Corresponding author: jenwils@uw.edu

using automated tools for performing translation.¹ identifying current translation practices, challenges, and needs, based on a survey of emergency managers across the United States. We then explore the translation tools available, and discuss the existing research on their evaluation and use in a crisis setting.

2 Establishing the Need: A Survey of Emergency Managers

We conducted a survey of members from the Language & Accessibility for Alert & Warning Workgroup (LAAWW) in December 2025. LAAWW is “a group of representatives from public and non-profit agencies working toward greater language and disability access in emergency alert and warning systems in the United States, U.S. Territories, and Tribal Nations.”² LAAWW has a broad and diverse membership consisting of approximately 360 members representing 42 US states and territories.

While LAAWW has broad national representation, this survey is not intended to be statistically representative of all U.S. emergency management agencies. Rather, it employs a purposive sampling strategy, targeting practitioners who are actively engaged in language and accessibility work. This method provides insight into current challenges, tools, and decision-making processes in multilingual alerting.³

The survey aims to understand the roles of LAAWW members in emergency management, languages spoken in their communities, translation tools and practices, and challenges faced in their efforts to communicate effectively with multilingual communities during crises. A total of 25 responses were received from across the country. Over 50% of respondents identified themselves as a *Local Emergency Manager* or *Public Information Officer*, and 36% identified as an *Operations, Logistics, or Communications Manager* or *Language Coordinator or Interpreter*.⁴

¹In this paper, we analyze commercially available tools that allow users to supply text and a target language, and we leave the analysis of MT workflows that require specific technical knowledge or an expertise in Machine Translation to others.

²<https://sites.google.com/view/laaww>

³This approach is appropriate for this paper’s goals: to document current practice, surface common challenges, and identify gaps. Future work with larger, randomly sampled populations would be needed to establish statistical representativeness.

⁴The survey allowed respondents to select more than one role.

Respondents represented cities as small as Chesterton, MD, where they support three languages and a population of 5,500, to New York City, where they support 200+ languages and a population of almost 8.5 million. Counties were also well represented, including Yuba County, CA, which supports 87,469 people and 3 languages, Guilford County, NC supporting 558,000 people and 100+ languages, and San Diego County, CA, supporting almost 3.3 million people and 9 languages. Taken together, LAAWW members who responded to the survey support over 36 million people living across the United States⁵.

Respondents noted that while many jurisdictions have residents speaking dozens or even hundreds of languages, only a subset of these languages can be consistently supported for time-critical emergency alerts, requiring agencies to make difficult prioritization decisions.

2.1 What are the translation needs across the LAAWW membership?

In 2023, the US Federal Communications Commission (FCC, 2023) recognized that “it is essential that the public be able to receive in accessible language and format WEA Messages that are intended for them...to help the millions of people with access and functional needs, including people who primarily speak a language other than English or Spanish...better understand and take protective actions...” Despite this policy emphasis, the ability of agencies to meet multilingual and accessible alerting needs varies widely depending on local demographics, staffing, funding, and available translation infrastructure. LAAWW members share the goal of transmitting accurate, actionable information to their communities in all relevant languages and take this responsibility seriously.

Translation needs are diverse and vary significantly across regions and agencies. According to the survey responses, the number of languages served ranges from as few as two to over two hundred, with most regions commonly including Spanish, though the specific languages can differ widely. The primary language for all regions is English.

In the United States, the national Wireless Emergency Alerts (WEA) service can send alerts to mo-

⁵The population figures reported here reflect the total populations of respondents’ jurisdictions and do not imply that all residents require translated alerts, nor that language services are uniformly provided across all languages in each jurisdiction. Rather, they illustrate the scale and diversity of contexts in which respondents operate.

mobile devices, along with other WEA-enabled devices, to provide actionable information in times of crisis, but the messages are sent in English. This leaves non-English speakers at a disadvantage unless they proactively sign up for translations.

Translations should not only be usable and accurate, but also be affordable and quickly produced. Although all respondents prioritized quality translations, for those who shared their challenges, *all* mentioned money, time, or limited resources (or all three) as challenges to meeting the needs of their communities.

2.2 What are the tools and methods used?

The majority (55%) conducted only human translation and did not use automated tools. Many used a mix of human translators and automated tools (27%) and some used only automated tools (18%). For those conducting human translation, work was carried out by organization staff (45%), community members or volunteers (26%), and contractors or commercial vendors (29%). This diversity highlights the effort involved in producing high quality translations.

Of respondents who report using automated tools, Google Translate⁶ was the choice in 90% of cases, indicating its central role in current translation workflows, with DeepL⁷ a distant second at 20%. A minority (and we suspect growing) number of respondents use Large Language Model (LLM)-based tools, with ChatGPT⁸ used in 30% of cases and Google Gemini⁹ used in 10%¹⁰. Given the small sample size, reported use of LLM-based tools likely reflects exploratory or supplemental use rather than standardized, organization-wide deployment.

2.3 Methods for Tool Selection

Selection of tools and methods appear to be driven primarily by practical considerations rather than formal requirements. Availability or ease of access is a selection criterion for 56% of respondents, with perceived accuracy or quality important for 52%, suggesting that organizations prioritize tools that are both readily usable and reliable. Cost and budgetary constraints are criteria for 40%, and ease of

use or staff comfort are important for 32%, further highlighting the impact of resource limitations and usability in everyday workflows. In contrast, compliance requirements (16%) and other factors (12%) were cited relatively infrequently, indicating that regulatory or external mandates play a smaller role in shaping tool choices. The relative infrequency with which compliance and formal requirements were cited suggests that translation tool adoption in crisis contexts is often driven by immediacy and feasibility rather than by systematic risk assessment or regulatory guidance.

2.4 Methods for Tool Evaluation

Evaluation methods are also driven by practical considerations. Only 10% of respondents indicate they have a formal evaluation process to examine how well these tools work for accuracy, usability, or effectiveness. The remaining respondents report they use an informal evaluation process (60%) or do not use one at all (30%).

Approaches to reviewing or updating translation tools vary across agencies and institutions. While 40% report reviewing or updating tools on an as-needed basis, a substantial share indicate limited formal review: 25% report having no formal review process, and an additional 25% are unsure whether such a process exists. A small proportion (10%) report that tools are not reviewed at all. These results indicate that, in many jurisdictions, translation tools are used in high-stakes emergency contexts without systematic evaluation of accuracy or effectiveness, raising concerns about uneven message quality across languages.

2.5 Languages in Need of Translation

LAAWW members in total have identified over 200 languages that their members are contending with. Within that number, there are a subset that were identified explicitly in the survey, some of which were mentioned repeatedly. Of particular note is Spanish, which was identified by almost all respondents who specifically listed languages that they needed to translate to and from. This is consistent with U.S. Census Bureau data (2024) showing Spanish is the most widely spoken non-English language in the United States, with over 44 million speakers.

2.6 Challenges Faced

Those performing translations for crisis communication face multiple challenges, including lim-

⁶<https://translate.google.com/>

⁷<https://www.deepl.com/en/translator>

⁸<https://chatgpt.com/>

⁹<https://gemini.google.com/app>

¹⁰ Respondents could specify more than one tool, which means the percentages do not necessarily add to 100%.

ited time, funding, staff, and support from management. Several respondents identified similar challenges regarding timely translations, wanting to “minimiz[e] time delays in alerts and warnings...especially with translations into various languages.” The “challenge [with] human translation is always time and availability” of translators. One respondent shared that their state “has a registry of approved interpreters with very few in our rural county.” Growing linguistic diversity requires support for additional languages, yet external vendor translations or human translators can be slow or unavailable. Identifying true language needs can be challenging.

Challenges for selection and evaluation of tools is also relevant to the efficacy of tools, with one respondent saying “multiple languages in the state are very unique thus singular translation services for only a few languages has proven ineffective in past.” Several noted that the usability of translations varied by language and tool used. Finally, it bears repeating that while all respondents prioritized quality translations, every respondent mentioned money, time, or limited resources (or all three) as a challenge to meeting the needs of their communities.

These constraints help explain the appeal of automated translation tools, while simultaneously underscoring the risks of deploying them without adequate evaluation, oversight, or language-specific quality assurances.

Several respondents also noted that language access does not guarantee message comprehension, particularly for populations with limited literacy or differing communication preferences. While the present survey focused on language translation, these findings point to the need for complementary accessibility strategies, such as plain-language messaging, audio or pictorial alerts, and community-based dissemination, especially in high-risk populations.

3 Measuring Translation Quality and Utility—A Problem in Itself

Measuring MT quality is not a trivial exercise. The gold standard for MT quality assessment is to use human evaluation. Human evaluators, however, do not come cheap, and even then, evaluations can have degrees of subjectivity, often dependent on the specific backgrounds of the evaluators and what instructions they might have been provided

(Han et al., 2021). The principal issue is that any given language is ambiguous, and translating expressions between languages only amplifies this ambiguity. For example, virtually any phrase or sentence can have multiple “correct” translations into another language. An example of this, presented at the International Workshop on Speech Translation (IWSLT) in 2004,¹¹ is the Japanese sentence: *ズキズキ 痛み ます*, about experiencing a throbbing pain. It was shown to have a minimum of 16 valid translations into English, all of which were created independently by Japanese translators, such as, *I have a throbbing pain*, *I am experiencing a throbbing pain*, *I am suffering from a throbbing pain*, *I am feeling a throbbing pain*, etc. This means that there is no one “correct” translation for this expression, which is true for almost any expression crosslingually. Multiplying this across a large, representative sample of sentences, say, one that is characteristic of crisis response, makes the evaluation problem seemingly intractable.

Despite this high degree of ambiguity, and subjectivity of potential evaluators, human evaluators can rely on their language skills, training and intuitions to arrive at a score for each sentence within a given test set. Then averaging that score with the scores for hundreds to thousands of other sentences within a test set, one can arrive at a reasonably accurate assessment of quality of an MT system (Chatzikoumi, 2020). If the test set is relevant to a particular domain, *e.g.*, crisis response, it can provide an assessment of the quality of the MT system for that domain. This kind of human evaluation is done frequently in large “bake-off” MT competitions such as the Conference on Machine Translation (WMT) (Lavie et al., 2025) and International Conference on Spoken Language Translation (IWSLT) (Salesky et al., 2025). But, as with human translation, human evaluators are also quite expensive, and MT research is often tasked to employ automated evaluation methods, especially for large-scale evaluation, multiple language evaluations, or during the development phase of MT. But how does an automated evaluator, *i.e.*, some sort of evaluation metric, arrive at an evaluation score when the input and output are so ambiguous? Therein lies the problem.

The most common method for evaluating MT

¹¹This example sentence and its translations were presented at the 2004 IWSLT (Akiba et al., 2004). The Japanese sentence itself was drawn from the Basic Travel Expressions Corpus (BTEC) (Takezawa et al., 2002).

quality is BLEU (Papineni et al., 2002). Developed at IBM in the late 90s, it was the gold standard for evaluation for many years, and is still one of the few options available for lower resource languages. Better methods exist, *e.g.*, BERTScore (Zhang et al., 2020), COMET (Rei et al., 2020), which do much better for modern translation algorithms, mostly because they are better at capturing the semantics of the output (BLEU relies much more heavily on string matching). However, these evaluation metrics only work for high resource languages, and are not available for those that are lower resource. For lower resource languages, there really are few options outside of BLEU.

BLEU measures the number of words and phrases that overlap between the a translation system’s outputs and one or more human reference translations. Just like with human evaluation, these scores are then averaged for all sentences within a test set. BLEU scores exist on a 0 to 1 scale, but are usually multiplied by 100 for reporting and easier interpretation. Higher BLEU scores signal better translation quality. A traditional rule-of-thumb to interpret BLEU scores is captured by Google’s MT API documentation¹² and shown in Table 1. Although measuring MT quality is the focus of much MT research, quality is not the only way to look at MT. Depending on the context, measuring *adequacy* (Specia et al., 2011) is also of interest. In MT, adequacy is usually used to indicate that a translation is suitable for the a user’s needs, whether or not it is fluent or grammatical. One can measure the adequacy, of an MT output in given contexts, such as utility of MT’d customer knowledge bases (Wendt, 2010)¹³ or even MT in the context of crisis response (Hajek et al., 2024).

In the following section we focus on the quality and utility of commercial MT tools in the crisis response context.

4 Quality of Commercial MT Tools in Crisis Response

The survey we conducted with LAAWW membership shows a clear interest in, and need for, auto-

BLEU Score	Interpretation
≥ 60	Quality often better than human
50 - 59	Very high quality, fluent translations
40 - 49	High quality translations
30 - 39	Understandable to good translations
20 - 29	Clear gist, major grammatical errors
10 - 19	Hard to get the gist
<10	Almost useless

Table 1: BLEU Score Interpretation

mated tools to support translation needs. While respondents indicate a strong preference for human translation, they also cite challenges surrounding resources, particularly availability of translators, cost, and the time taken to translate. The main commercial tool currently used by LAAWW members is Google Translate, though the survey also indicates use of AI tools including GPT, Gemini, and DeepL. However, many respondents expressed concerns about the accuracy of these tools, as well as a need for evaluation strategies, and for human verification of output translations.

To effectively assess the performance of commercial translation tools in crisis response, they need to be evaluated on domain specific content, specifically crisis-related content, and across a variety of languages—particularly languages that are the most needed. There are two recent studies that address this: Bansal et al. (2026), which is a quantitative evaluation of four machine translation systems when translating emergency pandemic-related content across 36 languages, and (Hajek et al., 2024), which is a more qualitative study across four languages.

4.1 Quantitative Evaluation

Bansal et al. (2026) is the first comprehensive, cross-lingual evaluation of commercial machine translation tools in the crisis domain that we are aware of. The authors evaluated four commercial systems—Google Translate, Microsoft Translator, GPT4o and Gemini 2.5—on pandemic-related content across 36 languages, using evaluation data drawn from the TICO-19 dataset (Anastasopoulos et al., 2020). For each of the 36 languages, they translated the evaluation sentences from English to the target language (EX) and from the language to English (XE). They then used this translation output and evaluated it against the human reference translations provided in TICO-19.

TICO-19 itself was created to facilitate the evaluation of a large set of high priority languages

¹²<https://cloud.google.com/translate/docs/advanced/automl-evaluate>

¹³Wendt (2010) looked at the differences between human translated content and machine translated content in Microsoft’s knowledge base, measuring adequacy on one criterion: “Did the knowledge base article you read satisfy your need?” Interestingly, when measured on this adequacy criterion, there was *no significant difference* between MT’d vs. human translated content, irrespective of target language.

spoken by communities most likely to be adversely affected by the COVID-19 pandemic.¹⁴ It contains 3,071 pandemic-related sentences from 38 languages, including 21 that are low-resource and spoken mostly in Africa and Southeast Asia, that Translators without Borders (TwB) labeled as *Priority*, based on predicted translation needs at the time. Additionally, there are 9 *Pivot* languages, each of which functions as a global or regional lingua franca and spoken by millions of people around the world, and 8 *Important* languages, with large speaker populations across Asia.

Although TICO-19 was focused on pandemic response, the fact that much of the content is medically-related, specifically *medically-related crisis content*, it is likely representative of content relevant to crises in general. Further, a subsection of TICO-19 is focused on *News and Announcements*, which may be particularly relevant to WEA-style alerts.

Bansal et al. (2026) focused not just on quality, but on the usability, of output translations. They define a translation as being usable if it is of sufficient quality (*good enough*) to be understood, and accurately conveys the relevant information (*adequate*). The translation does not need to be perfect to be usable. They considered languages with a BLEU score of at least 30 to be usable, and a score of 25-29 to be borderline (Table 1).

Based on the needs of LAAWW members, we focus on Bansal et al.’s (2026) results in the EX direction as they are the most relevant to WEA-style messages, which, as noted previously, are written in English, translated, then broadcast to affected populations. Additionally, the TICO-19 dataset consists of sentences taken from 6 sub-domains—*Conversational Medical*, *Scientific Medical*, *General* (but still COVID-related), *News*, *Announcements*, and *Travel*. Of these, we identified News and Announcements as the most relevant to the format of emergency alerts.

Of the languages identified by LAAWW members 19 of them overlap with those present in TICO-19 and are included in Bansal et al. (2026)’s evaluation. Table 2 presents the Overall EX BLEU scores for these languages, as well as additional scores for the News and Announcements (N+A) domain (calculated based on data shared by the authors). The full results for all languages can be found in Bansal

et al. (2026).

We see that *Pivot* languages are well supported by LLMs, Google Translate, and Microsoft Translator, and all have Overall scores above the 30 point threshold regardless of the translation tool used. For some languages, such as French and Chinese, various systems achieve scores above 40 and even 50 points, and are likely to produce good quality translations during real-time crisis scenarios. This pattern largely holds when looking specifically at the N+A domain, excepting Arabic and Hindi, where scores across models reflect at most borderline usability. This is particularly significant for Hindi, where each model shows a difference of over 10 points between N+A and Overall scores.

Languages labeled *Important* fared less well, with only Tagalog consistently scoring above 30, and Farsi scoring above the 25 threshold throughout. When looking at Overall scores for Urdu, Google produced borderline translations, while all other systems were unusable, and all systems were unusable for Myanmar and Tamil. However, in this case the N+A specific results are promising - Microsoft produces borderline usable translations for both Urdu and Tamil, while Gemini is also borderline for Urdu, leaving Myanmar as the only language without at least one usable system. Commercial systems do poorly on *Priority* languages, however—no system produced usable translations from English into these languages. The important exception is Microsoft Translator’s performance on Somali, which produced usable translations within the N+A domain, despite a low Overall score. Bansal et al. (2026) note that each system performs better when translating into English (XE), with around half of *Priority* languages achieving either usable or borderline scores. This is unfortunately not useful when translating emergency alerts from English.

When looking at Overall scores, Google consistently does the best across languages, regardless of resource level. The only exceptions are Arabic and Dari,¹⁵ where GPT-4o performed the best, and Tamil and Somali, where Microsoft performed the best. Focusing on the N+A domain results, however, we see more variation, specifically within *Pivot* languages, where Google was the best for only Portuguese, Chinese, and Hindi, while Microsoft was the best for Spanish, GPT-4o was the

¹⁴TICO-19 was created at the beginning of the pandemic, before it had reached all corners of the globe.

¹⁵Google Translate did not support Dari at the time data was collected.

Language	Overall BLEU Score				News and Announcements BLEU Score			
	MS	Google	GPT-4o	Gemini	MS	Google	GPT-4o	Gemini
——Pivot——								
Arabic	25.16	32.52	33.14	32.02	24.55	28.43	25.57	29.58
Portuguese (BR)	47.86	51.5	49.12	49.75	57.81	58.25	54.60	54.51
Chinese	38.29	52.59	48.61	44.38	33.95	48.28	46.95	38.23
French	40.35	44.57	43.12	40.92	35.32	37.30	54.39	49.44
Hindi	40.27	42.62	36.37	38.6	25.31	27.96	23.60	26.52
Spanish (LA)	51.33	56.92	48.56	48.56	60.66	59.27	56.51	57.72
Russian	32.94	37.30	32.35	30.48	35.38	34.96	36.17	39.10
Swahili	30.74	34.56	34.24	32.55	33.99	34.10	37.30	37.51
——Important——								
Farsi	25.25	29.75	25.89	25.55	25.46	28.01	29.53	26.16
Myanmar	1.25	8.03	4.66	4.41	0.94	6.57	5.06	4.14
Tagalog	49.4	49.86	39.46	38.38	48.65	50.36	43.56	38.85
Tamil	24.84	22.87	11.11	15.37	27.17	22.24	13.22	16.30
Urdu	20.41	27.69	23.07	23.65	28.43	29.85	24.59	26.30
——Priority——								
Amharic	7.04	11.13	5.85	10.14	8.05	9.77	4.23	6.78
Dari	15.21	—	18.46	17.95	15.26	—	19.21	17.90
Khmer	11.52	18.52	8.55	6.07	16.47	16.48	11.30	11.06
Nepali	14.20	24.30	18.39	18.75	12.25	19.28	17.89	16.29
Pashto	12.79	22.45	16.42	13.72	7.50	9.92	9.20	8.48
Somali	16.98	8.87	7.78	7.56	32.90	16.83	6.39	14.74

(Bansal et al., 2026)

Table 2: 2025 BLEU scores and usability of all four systems for each language when translating from English to X (EX), including overall scores and scores specifically for sentences within the News and Announcements domain. Being 'usable' means having a BLEU score of at least 30. Unusable systems are shaded with dark gray. Light gray indicates borderline usable systems. The best system for each direction is bold.

best for French, and Gemini was the best for Arabic, Russian, and Swahili. For *Important* and *Priority* languages, there is more consistency, the only exceptions being Farsi, where GPT-4o is the best system, and Khmer, where Microsoft scores the same as Google.

Finally, there are many cases where there is a large difference in Overall and N+A domain-based scores, indicating a substantial difference in the quality of output translations, even if the overall usability classification remains unchanged. This would impact the way a system is utilized in an emergency setting. For example, for Pashto, the N+A scores for Google were over 10 points lower than the Overall score, and between 5-10 points lower for the remaining systems. Conversely, for Spanish and Portuguese, 3 out of 4 models had N+A scores that were between 5-10 points higher than Overall scores. French was unusual, in that both LLMs had significantly higher N+A scores than Overall scores, while neural models had sig-

nificantly lower N+A scores. More generally, it appears that for *Pivot* languages, LLMs are competitive with neural models specifically within the N+A domain, but neural models, particularly Google Translate, are better options for *Important* and *Priority* languages.

This kind of variation in the N+A scores indicates that the style of the text, or the type of text being translated, has an impact on model performance; it is not dependent only on, for example, domain-specific vocabulary. Focusing specifically on N+A results then, we see that for most *Pivot* languages, all systems are usable, and for the remaining languages are at the least borderline. For most *Important* languages, at least one system exists that is borderline usable, but for *Priority* languages, commercial systems are still mostly unusable.

4.2 Qualitative Evaluation

Hajek et al. (2024) did a *qualitative* study on Machine Translation tools used in crisis response, in

addition to quantitative evaluations over human evaluator scores. They focused mostly on Google Translate, and the languages evaluated were Chinese, Dari, Greek, and Spanish, three of which overlap with languages identified by LAAWW members. It is worth taking a look at some examples from their qualitative analysis.

One notable example is the sentence “Try to position the car towards the approaching fire.” When this was translated back from Chinese into English the sentence was “Try to park your car close to a fire.” This is a dramatically different meaning, and could be potentially endangering to populations who receive this translation.¹⁶ Another example they give from Spanish is the sentence “Shelter in a room.” This translated into Spanish as “refugio en una habitación,” which when backtranslated, means “a refuge in a room.” The word “shelter” in this case was assumed to be a noun, when actually it was a verb (a command to take shelter).

These examples illustrate a type of MT failure that is dangerous in emergency contexts: mistranslations that are fluent, possibly plausible-sounding that therefore unlikely to be flagged by non-expert reviewers, but convey meaningfully different instructions (Šorak, 2024).

Qualitative evaluations can provide examples that show where MT has gone wrong. It can be useful in providing actionable content, which can help guide how translations are done, used, or evaluated. However, it does not give an overall picture of MT output, just specific instances identified by human translators. They may or may not be representative of broader issues. This limitation underscores the need for systematic, domain-specific evaluation frameworks for MT in emergency contexts.

5 What to do and what not to do with Commercial MT Tools

The gold standard for translations of emergency alerts and messaging is to translate all anticipated messages by a human before they are needed (Federici, 2022; Villarreal et al., 2025). When an organization has limited resources, or an unanticipated crisis occurs, or where professional translators are

not available, this process may not be possible. Commercial machine translation can bridge the gap and get information out to people quickly, while buying time to have a more thorough translation prepared.

Based on the work by Bansal et al. (2026), we can see that there are major gaps in the quality of commercial MT tools in the crisis domain cross-lingually, and potentially undesirable errors in output, such as those noted by (Hajek et al., 2024). Knowing where the quality gaps are can help emergency managers integrate translation tools into their workflows. Further, it is important to know how one can improve the quality of MT and how much a human should be in the loop. In this section, we will look at ideas and rules of thumb that emergency managers can adopt should they choose to use commercial translation tools.

5.1 Upsides and Downsides of Raw Machine Translation

Historically, it was considered poor form to use raw MT, *i.e.*, using the direct output of MT engines without any post-editing or human-in-the-loop, since the chance for error was high. As MT quality has improved in recent years, especially for high resource languages, the use of raw MT in a variety of settings has become more acceptable, especially in contexts where human translators are not available (Nurminen, 2021; Nurminen and Papula, 2018; Kasperé et al., 2021; Vieira et al., 2022; Kenny et al., 2022). This is the likely scenario that many LAAWW members find themselves in. However, there is a significant literature on the use of MT in crises, with the general recommendation that humans should be in the loop (Cadwell et al., 2019; O’Brien and Federici, 2019; Federici and Cadwell, 2018; Federici, 2022), mostly to prevent errors, which can sometimes be catastrophic and can adversely impact the lives of affected populations.

Avoiding the use of raw MT, however, may not be feasible when contending with dozens to hundreds of languages spoken in a region along with budget limits on the hiring of translators or editors, budgetary constraints being a common theme among the LAAWW members’ responses. Hajek et al. (2024) would seem support this, noting: “raw machine translations should not be used for emergency texts as English-language source texts are currently written. However, machine translations might be used as a last resort whenever adequate

¹⁶Hajek et al. (2024) gathered their translations in 2023. Upon examining Google Translate today (captured on 14 Jan, 2026), the Chinese translation is 着将汽向逼近的火焰。 which translates to “Try driving the car toward the approaching flames.” This is dramatically worse, especially in the context of a wildfire! It should be noted that this is not an issue with Google Translate alone; Microsoft Translator gives the back translation result of “Try to get as close to the fire as possible.”

preparations have been made, especially the writing of clear source texts, the prior translation of templates, the use of human post-editing when possible, and the identification of appropriately selected bilingual community-based contact persons.”

When raw translations are used without human review, accuracy is not guaranteed, increasing the risk to the population if incorrect information is provided. Organizations must carefully balance the risk and benefits of using this fast and inexpensive method. But there are things that can be done to help mitigate some of these dangers, either through simplifying text, preparing messages in advance, use of templates, and do as much as possible in advance of crises.

5.2 Translated Templates

The Federal Communication Commission (FCC) provides templates for WEA, which have been translated into the 13 most commonly spoken languages in the United States.¹⁷ By using templatic approaches like this, one can sidestep the use of MT altogether.

We recommend creating templates in advance for crises that may occur regularly. Use the following steps to create templates that can be repurposed (beyond those already provided by a governing body, such as the FCC):

- Create a set of pre-approved templates, using short, plain-language messages for hazards or actions. The templates can be created to address incidents that are likely to occur for each specific area. Examples from the FCC can be seen in Figure 1.
- Use clearly labeled and unambiguous placeholders in the templates for information that will only be known when a crisis happens (e.g., location and place names, dates, times, etc.). Ensure that the placeholders are not translated, and have the correct placement for the languages in question.
- When the relevant crisis situation happens and a message template has already been prepared, but not yet translated, it can be filled in and translated to additional languages through automated tools.

¹⁷<https://www.fcc.gov/multilingual-wireless-emergency-alerts>

English	[SENDING AGENCY]: A FLASH FLOOD EMERGENCY is in effect for [LOCATION] until [TIME]. This is an extremely dangerous and life-threatening situation. Do not attempt to travel unless you are fleeing an area that may flood or are under an evacuation order.
Chinese (Simplified)	[SENDING AGENCY]: 当前在[LOCATION]发布暴洪紧急状态, 持续生效至[TIME]。这是一种极其危险且危及生命的情况。请勿尝试旅行, 除非您要逃离可能遭受洪水侵袭或接到疏散令的地区。

Figure 1: Example of messaging templates (FCC 2025)

5.3 Text Simplification

Text simplification has been a traditional method of improving MT output, discussed in the literature under various guises: text simplification (Štajner and Popovic, 2016), controlled language (Aikawa et al., 2007), and pre-editing (Sánchez-Gijón, 2022), among others.

In the distributed messages setting, such as WEA, all messages should be as clear and direct as possible to increase the chance of information being translated correctly. Štajner and Popovic (2016) conclude that “the use of the [text simplification] can improve the fluency of the MT output...” While their work is based using automated simplification techniques, we argue that it can be broadened to include manual simplification techniques as well, i.e., where the authors of the messages simplify the text themselves. Hajek et al. (2024) suggest several methods to do this, including avoiding culture-specific terms, avoiding two-part verbs (e.g., look up, get along with, etc.), and avoiding elements with multiple grammatical functions. Many working in crisis response recommend message simplification before translation: local governments (King County Executive, 2010), policy reports (Fordham, 2023), and government reports (Hajek et al., 2024).

LLMs can also be used in text simplification (Guidroz et al., 2025). As an example, the following text could be used as a prompt to an LLM to simplify succeeding text supplied by the user:

Rewrite this text in plain language at approximately a 6th-grade reading level. Use short sentences and everyday vocabulary. Keep all safety instructions, locations, and times accurate.

The text simplification can be done as part of the translation step as well, if the language one wishes to translate into is supported by the LLM of choice. For example, one could rewrite the first sentence of the above prompt, changing “Rewrite” to “Translate into Spanish.” The rest of the prompt could remain the same. The LLM would then simplify the text *and* translate it in one step. See more about the use of LLMs in translation in the following

section.

5.4 Using LLMs for Translations

The bulk of LAAWW members use commercial translation tools, as noted earlier, but several respondents indicated an interest in using LLMs in translation. Although LLMs offer flexibility in how translations are done, the quality tends to be lower than existing translation tools and LLMs have limited language coverage. It is not always easy to discern what languages an LLM covers nor how well it will translate. Asking the LLM will not necessarily generate an honest response.

The use of general-purpose LLMs such as GPT and Gemini in high-stakes communication contexts introduces risks that go beyond translation quality as measured by BLEU scores. LLMs are known to produce fluent but semantically incorrect outputs — a phenomenon termed "hallucination" — at rates that are difficult to predict and vary by language, domain, and prompt formulation (Ji et al., 2023). Sriramanan et al. (2024) show that some LLMs have hallucination rates approaching 50%. In crisis communication, where a mistranslation may cause recipients to take dangerous rather than protective action, this unpredictability is a serious concern. Furthermore, agencies deploying MT tools without adequate human oversight may face legal exposure: Executive Order 13166 (2000) and Title VI of the Civil Rights Act require recipients of federal funding to provide meaningful language access, and failure to do so — particularly when a known-deficient tool is used without verification — could constitute a violation.¹⁸ The lack of formal evaluation processes reported by LAAWW respondents (only 10% have a formal evaluation process) amplifies this liability risk.

To minimize chances of hallucinations, the following or similar prompt could be used, which tells the LLM to leave text untranslated if there is any uncertainty. Although this prompt may reduce the chance of hallucination, there are no guarantees. For languages not well supported by an LLM (Table 2 can give some guidance), it is best to avoid their use.

Translate this message into [TARGET LANGUAGE]. Be literal. Do not add or invent any information. If a phrase is ambiguous, mark it with [UNCERTAIN] and leave it untranslated. Preserve

names, dates, and numbers exactly. After the translation, show a one-line confidence: High / Medium / Low and list any parts you were unsure about.

The authors acknowledge the mixed messaging this section provides. While we suggest that LLMs not be used for translations, we also give guidance on how to use LLMs for translations. As researchers, we are hesitant to recommend the use of LLMs for the reasons stated above. As people who have some understanding of the community that uses these tools, we know that LLMs may be used for translations regardless of what an academic paper may recommend. To that end, we also recommend strategies to use LLMs as effectively as possible.

5.5 Post-editing and Costs

It is challenging for organizations to determine where to best spend their limited resources. When it comes to translations, beyond the methods discussed previously (e.g., using templates, simplification, etc.) one option is to spend resources on post-editing rather than on message creation and translation. In this scenario, human translators will rely on MT to produce translated output and then correct any errors in the output. This allows translators to spend their time reviewing and editing translated messages rather than translating messages from scratch. Post-editing is often done using Computer Assisted Translation (CAT) tools such as Trados or MemoQ¹⁹, although it can be done in tools such as Word or Excel (with a little more difficulty). One of the advantages of using CAT tools is that they track translations and can reuse material from earlier translations (through *Translation Memories*). Many professional translators are familiar with these tools.

More ad hoc translations probably occur regularly in crisis situations, given the ubiquity and availability of translation tools and people's familiarity with them. In urgent situations, messages are often sent in dominant language(s) first before they are translated into other languages. For people who do not speak the dominant language, urgency will drive them to quickly obtain their own translations, using some of these same commercial translation tools described in this paper, rather than waiting for official, translated messages. In many cases, a translated message may not be available at all for a given language, and users are left on their own to

¹⁸This executive order was repealed by Executive Order 14224 on 01 March 2025. While 13166 is not currently in force, future administrative actions may put it back in place.

¹⁹<https://www.trados.com/>, <https://www.memoq.com/>

translate.

To ensure that the messages are broadly consumable, including translated where needed, we recommend:

- Follow the steps in the Translated Templates section. The steps will increase the readability, and translatability, of messages.
- Share the messages on all available public and social platforms. Some of these platforms allow users to quickly translate messages without pasting the text into a separate program or app.

Although this removes the ability to monitor quality of the translations and increases the risk that people will get bad, incorrect, or harmful information, it does increase the number of people reached.

5.6 Evaluating Translations

MT engines can produce errors, and sometimes these can be catastrophic (such as the examples from (Hajek et al., 2024)). It is strongly recommended that translations be evaluated whenever possible or feasible. Although quality evaluation is certainly possible, as reviewed earlier in the paper (and performed by Bansal et al. (2026)), it may be more useful to determine *adequacy* for the specific need versus absolute *quality*. In other words, a translation that has some limited errors or is ungrammatical, may still be of sufficient quality and good enough to be used. Here are some rules of thumb for determining the adequacy of the translations provided by a commercial translation tool (or of any translations):

- Use Multidimensional Quality Metric (MQM) (Lommel et al., 2024) to define what *major errors* might exist. Ultimately, there should be no *major errors*, but minor errors may be acceptable. Examples of major errors may be: incorrectly translated safety instructions, incorrect locations, or incorrect times.
- Critical data, such as place names, should be kept in the original language because geographic place names tend to stay consistent and do not change based on language spoken. This information can be evaluated even by those who do not read the target language.

In scenarios where time is limited or it is not possible to do a formal evaluation, one can do round-trip translation and cross checking. For round-trip translation, use an MT tool to perform the initial translation into the target language. Once the trans-

lation output is generated, copy the translated text into a different MT tool and translate it back into the original language. If the new translation preserves the meaning, then the translation is likely usable. To use cross checking, use two different MT tools to produce a translation for the same text and compare the outputs. If the outputs are identical and likewise pass the back translation test, then the translation is likely usable.

6 Conclusion and Future Work

We recommend having humans in the loop whenever possible, for translating, editing, and evaluation. But realistically, it is just not possible budget-wise to do so at all levels, especially in regions where translations need to be produced in dozens of languages. Automating workflows, wherever possible, is realistically the only option. Likewise, knowing to what level one might be able to automate is equally important to know.

An obvious, but important point to make: MT will make errors, and not all errors are created equally. In fact, the kinds and severity of errors can vary greatly depending on the kinds of content that we are trying to translate, *i.e.*, the complexity of the original text or the language that is being translated to. As we have shown, MT tends to do better with higher resource languages such as Chinese or Spanish, and less well for languages such as Amharic or Pashto. How one mitigates and plans for these errors can make a difference in high-stress, time-sensitive crisis scenarios. What is most important in the crisis context is that context-critical information that needs to be relayed, such as place names, dates, times, etc., be provided even if the underlying translation is ungrammatical or not fluent.

It is also important to have MT evaluation as part of any workflow. Inspecting the quality or adequacy of MT output, which we have shown can vary widely by language, can ensure that MT satisfies the needs of affected communities.

In a world where communities are becoming ever more multilingual, communication is increasingly relying on translation. Fortunately, the quality and utility of MT has only increased over the years, making it a viable solution for communication, even in time-sensitive crisis scenarios.

Limitations

The survey done for this paper captures the languages that agencies *report* needing to support. It does not systematically compare those responses against actual distribution of non-English speakers in the respondents' jurisdictions as measured by US Census data or available American Community Survey data.

The fact that there are few options to evaluate lower resource languages outside of BLEU is a significant constraint in the context of this paper. Many of the languages most in need of support are precisely those for which advanced evaluation metrics are unavailable, leaving agencies with limited tools to verify translation quality and usability.

References

- Takako Aikawa, Lee Schwartz, Ronit King, Mo Corson-Oliver, and Carmen Lozano. 2007. [Impact of Controlled Language on Translation Quality and Post-editing in a Statistical Machine Translation Environment](#). In *Proceedings of the Machine Translation Summit XI*.
- Yasuhiro Akiba, Marcello Federico, Noriko Kando, Hiromi Nakaiwa, Michael Paul, and Jun'ichi Tsujii. 2004. [Overview of the IWSLT evaluation campaign](#). In *Proceedings of the 1st International Workshop on Spoken Language Translation: Evaluation Campaign*, Kyoto, Japan.
- Antonios Anastasopoulos, Alessandro Cattelan, Zi-Yi Dou, Marcello Federico, Christian Federmann, Dmitry Genzel, Francisco Guzmán, Junjie Hu, Macduff Hughes, Philipp Koehn, Rosie Lazar, Will Lewis, Graham Neubig, Mengmeng Niu, Alp Öktem, Eric Paquin, Grace Tang, and Sylwia Tur. 2020. [TICO-19: the translation initiative for COvid-19](#). In *Proceedings of the 1st Workshop on NLP for COVID-19 (Part 2) at EMNLP 2020*, Online. Association for Computational Linguistics.
- Vipasha Bansal, Elizabeth Brown, Chelsea Kendrick, Benjamin Pong, and William D. Lewis. 2026. Is MT Ready for the Next Crisis or Pandemic? <https://arxiv.org/abs/2601.10082>. arXiv:2601.10082.
- U.S. Census Bureau. 2024. [Language spoken at home](#). U.S. Census Bureau. Accessed on 22 April 2026.
- Patrick Cadwell, Sharon O'Brien, and Eric DeLuca. 2019. More than tweets: A critical reflection on developing and testing crisis machine translation technology. *Translation Spaces*, 8(2):300–333.
- Eirini Chatzikoumi. 2020. How to evaluate machine translation: A review of automated and human metrics. *Natural Language Engineering*, (26).
- FCC. 2023. [Emergency Alert System; Wireless Emergency Alerts. Final Rule](#).
- Federal Communications Commission. 2025. [Multilingual wireless emergency alerts](https://www.fcc.gov/multilingual-wireless-emergency-alerts). <https://www.fcc.gov/multilingual-wireless-emergency-alerts>. [Accessed 14-01-2026].
- Federico Federici and Patrick Cadwell. 2018. Training citizen translators: Design and delivery of bespoke training on the fundamentals of translation for New Zealand Red Cross. *Translation Spaces*, 7(1):20–43.
- Federico M. Federici. 2022. [Translating hazards: multilingual concerns in risk and emergency communication](#). *The Translator*, 28(4):375–398.
- Anni Fordham. 2023. Beyond barriers. <https://churchilltrust.my.salesforce-sites.com/api/services/apexrest/v1/image/?Id=069RF0000NxDTtYAN&forceDownload=Yes>. [Accessed 06-01-2026].
- Theo Guidroz, Diego Ardila, Jimmy Li, Adam Mansour, Paul Jhun, Nina Gonzalez, Xiang Ji, Mike Sanchez, Sujay Kakarmath, Mathias MJ Bellaiche, Miguel Ángel Garrido, Faruk Ahmed, Divyansh Choudhary, Jay Hartford, Chenwei Xu, Henry Javier Serrano Echeverria, Yifan Wang, Jeff Shaffer, Eric (Yifan) Cao, and 7 others. 2025. [LLM-based Text Simplification and its Effect on user Comprehension and Cognitive Load](#). arXiv:2505.01980.
- John Hajek, Anthony Pym, Yu Hao, Maria Kariadakis, Ambrin Hasnain, Anila Hasnain, Juerong Qiu, Ke Hu, and Rachel Macreadie. 2024. Understanding and improving machine translations for emergency communications. <https://works.hcommons.org/records/j3fba-nsn06>. [Accessed 06-01-2026].
- Lifeng Han, Gareth Jones, and Alan Smeaton. 2021. Translation quality assessment: a brief survey on manual and automatic methods. In *Proceedings of MoTra21: Workshop on Modelling Translation: Translatology in the Digital Age*, Reykjavik.
- Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. 2023. [Survey of hallucination in natural language generation](#). *ACM Comput. Surv.*, 55(12).
- Ramunė Kasperė, Jolita Horbačiauskienė, Jurgita Motiejūnienė, Vilmantė Liubinienė, Irena Patašienė, and Martynas Patašius. 2021. [Towards Sustainable Use of Machine Translation: Usability and Perceived Quality from the End-User Perspective](#). *Sustainability*, 13(23):13430.
- Dorothy Kenny, Olga Torres-Hostench, Caroline Rossi, Alice Carré, Pilar Sánchez-Gijón, Sharon O'Brien, Joss Moorkens, Juan Antonio Pérez-Ortiz, Mikel L. Forcada, Felipe Sánchez-Martínez, and Gema Ramírez-Sánchez. 2022. [Machine Translation for Everyone](#). Language Science Press.

- King County Executive. 2010. Written language translation process king county written language translation process manual guidelines, effective practices, maps, resources. <https://kingcounty.gov/exec/styleguide/~media/cbf01a3bf9644c69b9ca5af88abb1202.ashx?la=en>. [Accessed 07-01-2026].
- Alon Lavie, Greg Hanneman, Sweta Agrawal, Diptesh Kanojia, Chi kiu Lo, Vilém Zouhar, Frederic Blain, Chrysoula Zerva, Eleftherios Avramidis, Sourabh Dattatray Deoghare, Archchana Sindhuja, Jiayi Wang, David Ifeoluwa Adelani, Brian Thompson, Tom Kocmi, Markus Freitag, and Daniel Deutsch. 2025. [Findings of the WMT25 Shared Task on Automated Translation Evaluation Systems: Linguistic Diversity is Challenging and References Still Help](#). In *Proceedings of the Proceedings of the Tenth Conference on Machine Translation (WMT25)*, Suzhou, China.
- Arle Lommel, Serge Gladkoff, Alan Melby, Sue Ellen Wright, Ingemar Strandvik, Katerina Gasova, Angelika Vaasa, Andy Benzo, Romina Marazzato Sparano, Monica Foresi, Johani Innis, Lifeng Han, and Goran Nenadic. 2024. [The multi-range theory of translation quality measurement: Mqm scoring models and statistical quality control](#). Preprint, arXiv:2405.16969.
- Patrick Meier. 2015. *Digital Humanitarians*. CRC Press, Boca Raton.
- Mary Nurminen. 2021. *Investigating the influence of context in the use and reception of raw machine translation*. PhD, Tampere University.
- Mary Nurminen and Niko Papula. 2018. Gist MT Users: A Snapshot of the Use and Users of One Online MT Tool. In *Proceedings of the 21st Annual Conference of the European Association for Machine Translation*, pages 219–228, Alicante, Spain.
- Sharon O’Brien and Federico Federici. 2019. Crisis translation: Considering language needs in multilingual disaster settings. *Disaster Prevention and Management*, 29(2):129–143.
- Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. 2002. Bleu: a method for automatic evaluation of machine translation. In *Proceedings of the 40th annual meeting of the Association for Computational Linguistics*, pages 311–318.
- Ricardo Rei, Craig Stewart, Ana C Farinha, and Alon Lavie. 2020. [COMET: A neural framework for MT evaluation](#). In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 2685–2702, Online. Association for Computational Linguistics.
- Elizabeth Salesky, Marcello Federico, and Antonis Anastasopoulos, editors. 2025. [Proceedings of the 22nd International Conference on Spoken Language Translation \(IWSLT 2025\)](#). Association for Computational Linguistics, Vienna, Austria (in-person and online).
- Pilar Sánchez-Gijón. 2022. Selecting and preparing texts for machine translation: Pre-editing and writing for a global audience.
- Vanessa Šorak. 2024. [From terminological inconsistencies to major mistranslations: A qualitative analysis of nmt errors in public health communication](#). *AI-Linguistica. Linguistic Studies on AI-Generated Texts and Discourses*, 1(1).
- Lucia Specia, Najeh Hajlaoui, Catalina Hallett, and Wilker Aziz. 2011. [Predicting machine translation adequacy](#). In *Proceedings of Machine Translation Summit XIII: Papers*, Xiamen, China.
- Gaurang Sriramanan, Siddhant Bharti, Vinu Sankar Sadasivan, Shoumik Saha, Priyatham Kattakinda, and Soheil Feizi. 2024. [Llm-check: Investigating detection of hallucinations in large language models](#). In *Advances in Neural Information Processing Systems*, volume 37, pages 34188–34216. Curran Associates, Inc.
- Sanja Štajner and Maja Popovic. 2016. [Can text simplification help machine translation?](#) In *Proceedings of the 19th Annual Conference of the European Association for Machine Translation*, pages 230–242.
- Toshiyuki Takezawa, Eiichiro Sumita, Fumiaki Sugaya, Hirofumi Yamamoto, and Seiichi Yamamoto. 2002. [Toward a Broad-coverage Bilingual Corpus for Speech Translation of Travel Conversations in the Real World](#). In *Proceedings of the Third International Conference on Language Resources and Evaluation (LREC)*, Las Palmas, Canary Islands.
- Lucas Nunes Vieira, Carol O’Sullivan, Xiaochun Zhang, and Minako O’Hagan. 2022. [Machine translation in society: Insights from UK users](#). *Language Resources and Evaluation*.
- Melissa Villarreal, Carson MacPherson-Krutsky, and Mary Angelica Painter. 2025. [Barriers and best practices for inclusive emergency alerts and warnings](#). *International Journal of Disaster Risk Reduction*, 125:105581.
- Chris Wendt. 2010. [Better translations with user collaboration - Integrated MT at Microsoft](#). In *Proceedings of AMTA 2010*.
- Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q. Weinberger, and Yoav Artzi. 2020. [Bertscore: Evaluating text generation with bert](#). In *International Conference on Learning Representations*.

Appendix

Language	English to X				X to English			
	MS	Google	GPT-4o	Gemini	MS	Google	GPT-4o	Gemini
——Pivot——								
Arabic*	25.16	32.52	33.14	32.02	44.87	48.14	43.7	43.70
Portuguese (BR)*	47.86	51.5	49.12	49.75	55.33	58.14	52.88	52.70
Chinese*	38.29	52.59	48.61	44.38	36.48	40.77	34.28	33.38
French*	40.35	44.57	43.12	40.92	45.32	45.20	47.42	49.91
Hindi*	40.27	42.62	36.37	38.6	53.28	59.35	46.97	48.97
Indonesian	46.85	54.44	53.56	51.35	50.29	53.78	49.69	51.67
Spanish (LA)*	51.33	56.92	48.56	48.56	54.59	54.57	49.07	51.32
Russian*	32.94	37.30	32.35	30.48	36.84	39.02	35.19	38.64
Swahili*	30.74	34.56	34.24	32.55	41.92	49.32	41.88	44.26
——Important——								
Bengali	17.18	20.3	15.96	13.18	46.08	49.95	43.12	43.99
Farsi*	25.25	29.75	25.89	25.55	39.35	43.35	37.80	40.28
Malay	45.78	52.22	45.23	49.19	54.08	59.00	50.76	52.22
Marathi	13.53	15.63	15.16	12.17	37.34	43.36	37.82	38.04
Myanmar*	1.25	8.03	4.66	4.41	31.44	37.98	27.63	37.95
Tagalog*	49.4	49.86	39.46	38.38	59.92	66.23	53.5	30.62
Tamil*	24.84	22.87	11.11	15.37	40.73	44.33	32.91	37.94
Urdu*	20.41	27.69	23.07	23.65	35.88	44.34	36.17	38.38
——Priority——								
Amharic*	7.04	11.13	5.85	10.14	33.02	37.76	28.51	37.52
Dari*	15.21	—	18.46	17.95	31.22	—	35.02	38.4
Dinka	—	4.38	—	—	—	14.65	—	—
Eritrean Tigrinya	3.10	6.17	—	4.54	18.00	29.86	27.48	28.55
Ethiopian Tigrinya	3.23	6.83	—	4.93	21.52	29.86	27.46	28.55
Hausa	25.23	27.43	22.44	20.4	33.92	39.70	32.97	36.08
Kanuri	—	4.74	—	—	—	26.19	—	—
Khmer*	11.52	18.52	8.55	6.07	37.04	46.44	36.88	41.01
Kinyarwanda	18.75	12.29	12.13	12.55	25.39	29.40	25.10	26.51
Kurdish Kurmanji	11.82	14.79	8.43	12.02	28.17	41.83	29.03	38.24
Kurdish Sorani	10.38	10.40	5.47	7.71	31.87	33.44	26.31	33.64
Lingala	28.35	14.70	12.99	14.52	29.14	25.64	22.44	25.45
Luganda	17.26	21.61	13.17	11.38	30.52	34.64	26.57	30.44
Nepali*	14.20	24.30	18.39	18.75	48.38	56.27	44.59	47.95
Nigerian Fulfulde	—	3.32	—	—	—	26.38	—	—
Oromo	—	11.58	6.41	7.56	—	32.04	25.95	29.56
Pashto*	12.79	22.45	16.42	13.72	32.16	43.9	35.03	37.22
Somali*	16.98	8.87	7.78	7.56	17.66	18.69	15.68	15.28
Zulu	15.92	19.57	15.69	16.55	43.40	50.88	40.79	41.9

(Bansal et al., 2026)

Table 3: 2025 BLEU scores and usability of all four systems for each language in either translation direction. Being “usable” means having a BLEU score of at least 30. Unusable systems are shaded with dark gray. Light gray indicates borderline usable systems. The best system for each direction is bold. The languages labeled with asterisks (“*”) are the same as those contained in Table 2 (EX only).